Time Series Event Detection in R: Anomalies, Change Points, Motifs

A comprehensive platform that brings together multiple detection methodologies under a single, coherent architecture — enabling systematic analysis of temporal patterns at scale.

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An Integrated Solution

Harbinger addresses a critical gap in time series analysis by providing a unified environment for anomaly detection, change point identification, and motif discovery — all within a single, coherent framework.

Available as an open-source R package, it empowers reproducible research and practical applications across a wide range of domains.

Harbinger

Stars 22 downloads 18K/month

Harbinger is a framework for event detection in time series. It provides an integrated environment for anomaly detection, change point detection, and motif discovery. Harbinger offers a broad range of methods and functions for plotting and evaluating detected events.

For anomaly detection, methods are based on: - Machine learning model deviation: Conv1D, ELM, MLP, LSTM, Random Regression Forest, and SVM - Classification models: Decision Tree, KNN, MLP, Naive Bayes, Random Forest, and SVM - Clustering: k-means and DTW - Statistical techniques: ARIMA, FBIAD, GARCH

For change point detection, Harbinger includes: - Linear regression, ARIMA, ETS, and GARCH-based approaches - Classic methods such as AMOC, ChowTest, Binary Segmentation (BinSeg), GFT, and PELT

For motif discovery, it provides: - Methods based on Hashing and Matrix Profile

Harbinger also supports **multivariate time series analysis** and **event evaluation** using both traditional and soft computing metrics.

The architecture of Harbinger is based on **Experiment Lines** and is built on top of the DAL Toolbox. This design makes it easy to extend and integrate new methods into the framework.

Why a Unified Framework?

Fragmentation

Specialized methods scattered across disparate tools and inconsistent interfaces create real barriers to practical analysis.

Real-World Demands

Modern applications require systematic comparison of detectors, seamless combination of methods, and rigorous evaluation — all in one place.

The Harbinger Solution

A coherent ecosystem where methods can be applied, evaluated, and combined consistently — turning complexity into clarity.

50+

Detection Methods

Comprehensive algorithm coverage

15+

Change Point Techniques

Approaches for structural breaks

10+

Motif Discovery Methods

Pattern recognition algorithms

Modular Architecture

Design Principles

The framework is built around a strict interface grounded in workflow algebra and experiment pipelines — ensuring consistency and interoperability across all detection methods.

Extending the foundation of the DAL Toolbox, it adopts design patterns inspired by the Scikit-learn API. The familiar `fit()` and `detect()` functions provide an intuitive, low-friction interface for researchers and practitioners alike.

01

Normalize

Preprocessing and standardization

02

Filter

Noise reduction and signal enhancement

03

Fit

Model training and parameter estimation

04

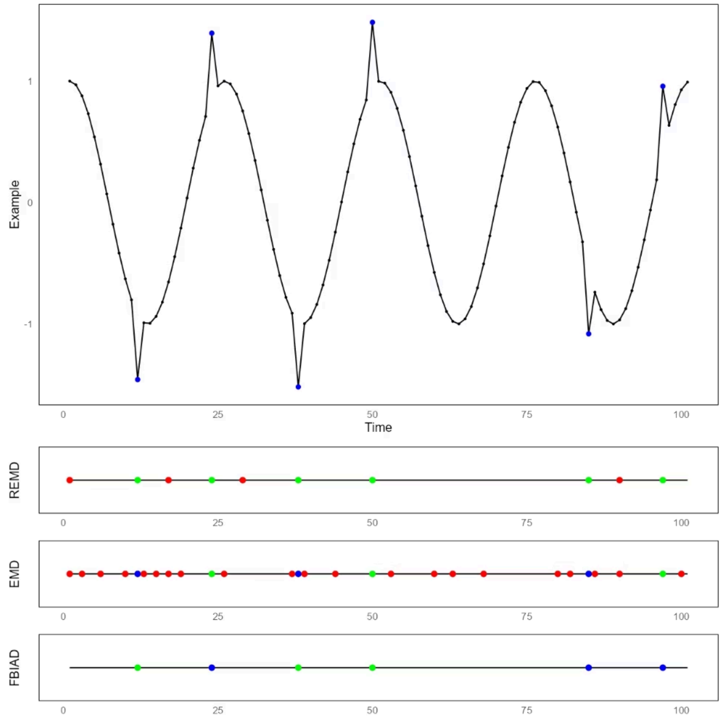
Detect

Event identification and classification

Detector Evaluation & Benchmarking

Harbinger offers comprehensive evaluation capabilities, enabling systematic benchmarking across diverse datasets and detection scenarios.

By implementing standardized metrics that quantify precision, recall, and temporal alignment, it enables objective, apples-to-apples comparison of competing algorithms.



Precision & Recall

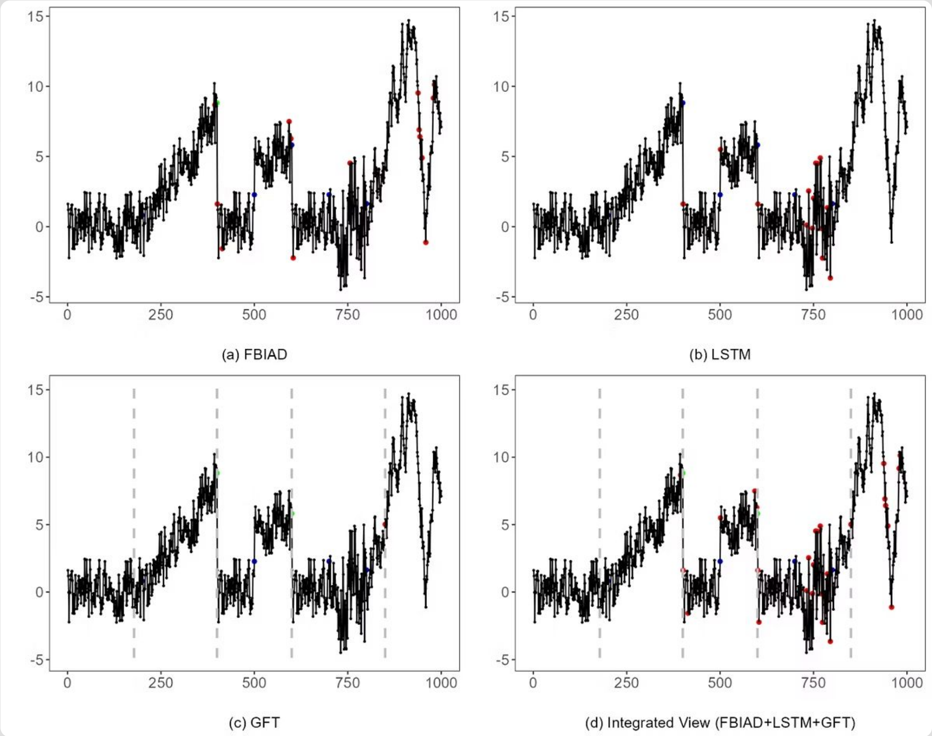
Quantify detection accuracy

Temporal Alignment

Measure timing precision

Performance Metrics

Comprehensive end-to-end assessment



Combining Detectors: Greater Than the Sum of Their Parts

Running multiple detectors simultaneously extends coverage, because each algorithm is sensitive to different patterns and helps suppress false positives. Rather than relying on a single perspective, Harbinger lets you combine signals to build a richer, more robust picture of what is happening across the time series.

When several methods converge on the same event, a **detector consensus** emerges — a strong indicator of high reliability and reduced ambiguity. This approach is especially valuable in critical domains such as healthcare, infrastructure, and finance, where decisions must be both defensible and precise.

Broader Coverage

Different detectors capture failures, patterns, and variations that any single method might overlook.

Consensus with Greater Confidence

Events confirmed by multiple algorithms carry stronger interpretive weight, making it easier to separate genuine signals from noise.

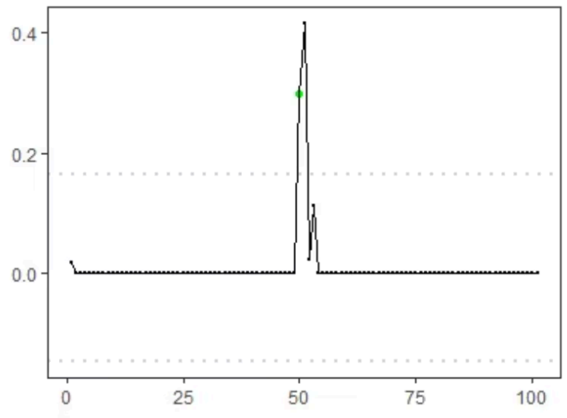
A Strategy Built for Critical Use Cases

Harbinger makes ensemble detection transparent and interpretable, supporting rigorous analysis in contexts where accuracy is non-negotiable.

Why Rigorous Evaluation Matters

In time series event detection, observing model outputs simply isn't enough. Without proper evaluation, we're left guessing about real performance. We need to quantify precision and recall to understand what a detector is genuinely capturing — and what it's missing entirely.

In practice, hard and soft evaluation serve distinct needs. In time series data, an event detected with a slight temporal offset can still be valuable — as long as there's an appropriate tolerance window. That's why comparing detectors consistently demands standardized metrics; only then does the analysis move from subjective impression to objective insight.



From Signal to Evidence

An observed detection doesn't prove quality — evaluation transforms raw outputs into measurable, defensible evidence.

Temporal Tolerance

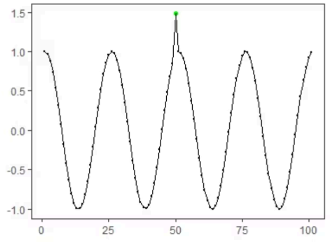
In time series contexts, small offsets are often acceptable when the goal is to catch the right event at approximately the right moment.

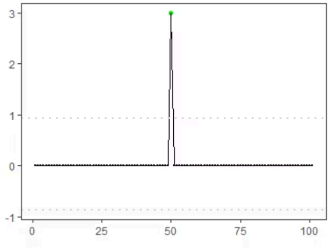
Fair Comparison

With standardized metrics, different detectors can be benchmarked systematically and reproduced consistently across analyses.

Standardized evaluation is part of Harbinger's core contract: **har_eval()** and **har_eval_soft()** ensure comparability, rigor, and consistency across all detection approaches.

Ensemble Models





Combining Multiple Detectors

Individual algorithms often exhibit complementary strengths and weaknesses. Ensemble methods bring multiple detectors together, leveraging their diversity to achieve more robust and reliable event identification.

```
model <- har_ensemble(
  hanr_fbiad(),
  hanr_arima(),
  hanr_emd())
model <- fit(model, dataset$serie)
detection <- detect(model, dataset$serie)
har_plot(model, dataset$serie,
  detection, dataset$event)
```

FBIAD

Frequency-based detection

ARIMA

Statistical modeling

EMD

Signal decomposition

Ensemble Aggregation

Combined decision output

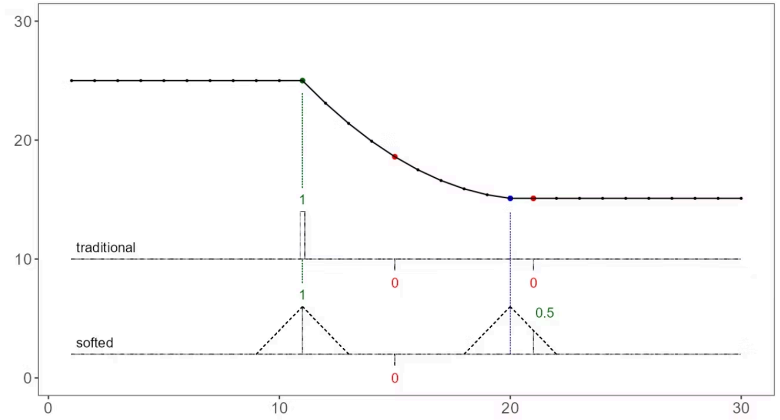
Fuzzifying Events: Beyond Binary Detection

SoftED Metrics

SoftED (Soft Event Detection) introduces fuzzy evaluation metrics that more faithfully capture how humans judge the quality of event detection.

Unlike traditional metrics that penalize any temporal misalignment equally, SoftED tolerates minor timing offsets while applying stronger penalties to fragmented detections — striking a balance that better reflects real-world expectations.

Fuzzy membership functions provide a graduated scale of detection quality, enabling a far more nuanced assessment of detector performance than a simple pass-or-fail verdict.



1

Temporal Tolerance

Accepts small timing misalignments without penalizing

2

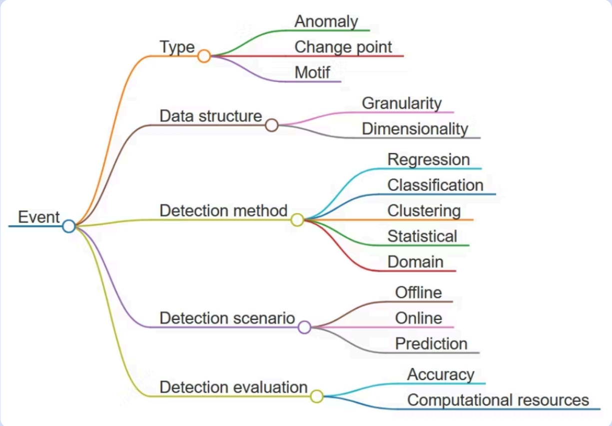
Fragmentation Penalty

Penalizes broken or split detections

3

Graduated Scoring

Delivers nuanced, continuous quality assessment



A Taxonomy of Time Series Event Detection

The field of time series event detection encompasses a rich diversity of methods, each designed to identify specific types of temporal patterns across vastly different domains.

This taxonomy organizes detection methods by the kinds of events they target and by their underlying algorithmic approaches — providing a clear map of the landscape.

Anomaly Detection

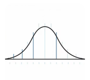
Identifies observations that deviate significantly from expected behavioral patterns.

Change Point Detection

Pinpoints moments when statistical properties undergo structural shifts.

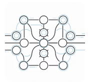
Motif Discovery

Uncovers recurring patterns or subsequences embedded within time series data.



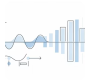
Statistical Methods

Leverage probabilistic models to characterize normal behavior and flag deviations



Machine Learning Approaches

Apply supervised or unsupervised learning to discover complex patterns in data



Signal Processing Techniques

Use frequency-domain analysis and decomposition to isolate distinctive event signatures

Hybrid Methods

Combine multiple paradigms to harness complementary strengths for greater accuracy

Summary

Unified Environment

Harbinger integrates anomaly detection, change point detection, and motif discovery.

Modular Architecture

Flexible composition of detectors and robust ensemble models.

Rigorous Evaluation

Precision/recall metrics ensure reliable and trustworthy results.

Fuzzy Event Representation

Beyond binary, for more nuanced and human-like analysis.

Open-Source & CRAN

Ready for community use, contributions, and active development.

Additional Sources

Explore the resources below to dive deeper into Harbinger and time series event detection.

CRAN Package

cran.r-project.org/web/packages/harbinger

Event Detection Resources

eic.cefet-rj.br/~eogasawara/en/event-detection-in-time-series

DAL Toolbox

cran.r-project.org/web/packages/daltoolbox

Issues & Discussions

github.com/cefet-rj-dal/harbinger/issues

Open questions, report bugs, or suggest improvements directly in the repository.